

Project Questions

Q1: Why Trust, Reasonableness, and Actionability?

The Core Argument

Standard AI metrics like accuracy, precision, recall — they measure if the model is correct. But in a Cyber-Physical System (power grid, water plant), that's not enough. An operator needs to know:

"Do I trust this explanation? Does it make sense to me? And what do I actually DO now?"

That's exactly what these 3 metrics capture.

Each Metric Explained

Trust — "Can I rely on this AI?"

In safety-critical systems, an operator will ignore an AI they don't trust, no matter how correct it is. If the AI says "mechanical failure" but there's clearly a storm outside, the operator loses trust immediately. Trust measures whether the explanation feels credible and consistent with reality.



Defend: CPS failures can cause blackouts, floods. Operators need confidence before acting. Trust is the foundation of human-AI collaboration in high-stakes environments.

Reasonableness — "Does this explanation make logical sense?"

An explanation can be technically correct but logically unreasonable to a domain expert. Example: AI says "pump failure" but sensor shows the storm started 2 minutes before the fault — a 20-year engineer knows the storm caused it. Reasonableness measures whether the explanation aligns with domain knowledge.



Defend: Traditional XAI (like SHAP) gives you feature importance numbers — not human-readable reasoning. We need a metric that checks if the logic of the explanation holds up to expert scrutiny.

Actionability — "What do I actually do now?"

An explanation is useless if it doesn't help the operator take action. "Sensors are abnormal" tells you nothing. "Lightning storm caused voltage sag — activate backup transformer and alert linemen" tells you exactly what to do. Actionability measures whether the explanation guides a concrete next step.



Defend: The whole point of XAI in CPS is to assist human decision-making under time pressure. If the output doesn't lead to action, the system has failed its purpose.

Why Not Standard Metrics?

Standard Metric	Problem
Accuracy/F1	Measures model correctness, not explanation quality
SHAP/LIME scores	Gives numbers (feature weights), not human reasoning
AUC-ROC	Classification performance — irrelevant for explanation evaluation

These three metrics evaluate the explanation's usefulness to a human, which is the entire goal of XAI.

Q2: How Does Agent 1 Detect Anomalies and Explain Them?

Step 1 — Anomaly Detection (Rule-Based Thresholds)

Before any agent runs, the system flags anomalies using hard engineering thresholds:

- Power Grid: Voltage < 110 kV AND Frequency < 49.5 Hz → anomaly flagged
- Water Plant: Pressure < 35 psi **AND** Vibration > 4.5 mm/s → anomaly flagged

This is intentional — in real CPS, engineers define safe operating ranges. We replicate that.



I Invigilator Q: "Why not ML for detection?"
I Rule-based thresholds are standard in industrial SCADA systems. Our contribution is not detection — it's explanation. The AI agents explain why the anomaly happened, not just that it happened.

Step 2 — Agent 1 Actually Has Two Sub-Agents

Agent 1a — Context-Agnostic Explainer (Traditional XAI)

- Receives only sensor values (voltage, current, frequency)
- Assumes mechanical failure by default
- Generates explanation without looking at external context
- Represents how traditional XAI systems work today

Agent 1b — Context-Aware Explainer (Our Proposed System)

- Receives sensor values plus External Context (storm, heatwave) plus Network Latency
- Links sensor deviations to external triggers
- Generates a richer, causally-grounded explanation

What Algorithm Runs Behind the Scenes?

Layer	Technology
Agent Framework	Microsoft AutoGen — manages multi-agent conversation
LLM (Brain)	LLaMA 3.2 running locally via Ollama
Agent Pattern	AssistantAgent (gives explanation) + UserProxyAgent (triggers it)

Layer	Technology
Detection	Rule-based threshold logic (Python/Pandas)

No SHAP. No LIME. No neural anomaly detector.

The "algorithm" is prompt engineering + LLM reasoning — Agent 1 is given a carefully designed system prompt that tells it its role, what data to use, and what format to reply in. The LLM then generates a natural language explanation.



I Invigilator Q: "Is this really XAI if it's just prompting an LLM?"

I Yes — because we are making the reasoning process explicit and transparent. The agent explains causality in plain language that an operator

I can verify and act on. That is the definition of explainable AI: not a black box score, but a human-understandable reason.

Complete Flow (One Sentence)

I User selects an anomaly → rule-based detection flagged it → Agent 1a explains it without context → Agent 1b explains it with full context →

I Agent 2 (expert evaluator persona) scores both on Trust, Reasonableness, Actionability → side-by-side comparison shows our framework wins.

XAI-CPS PROJECT — INVIGILATOR Q&A

Q1: Why did we choose Trust, Reasonableness, and Actionability as our model metrics?

THE CORE ARGUMENT

Standard AI metrics like accuracy, precision, recall measure if the model is correct. But in a Cyber-Physical System (power grid, water plant), that is not enough. An operator needs to know:

"Do I trust this explanation? Does it make sense to me?

And what do I actually DO now?"

That is exactly what these 3 metrics capture.

EACH METRIC EXPLAINED

TRUST — "Can I rely on this AI?"

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REASONABLENESS — "Does this explanation make logical sense?"

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WHY NOT STANDARD METRICS?

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Q2: How does Agent 1 detect anomalies and provide explanations? What algorithm is used behind the scenes?

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